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CS455 Computer Networks
Lab 3
September 11, 2026

1) Setup

I am using MacOs version Tahoe 26.6.2, I did not need to use WSL, and DiG version 9.10.6.

2) Dig Anatomy

```
; <<>> DiG 9.10.6 <<>> wikipedia.org A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2876
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0,
ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;wikipedia.org.         IN  A

;; ANSWER SECTION:
wikipedia.org.      96  IN  A   208.80.154.224

;; Query time: 4 msec
;; SERVER: 128.153.5.248#53(128.153.5.248)
;; WHEN: Mon Sep 07 14:55:14 EDT 2026
;; MSG SIZE  rcvd: 58
```

The output listed above was from running dig wikipedia.org A. The annotations that are made above are to denote the important sections listed in the lab instructions.

This color was used to denote the status of the query.
This color was used to denote the flags set

These flags mean:

- qr = response
- ra = recursion available
- rd = recursion desired

This color was used to denote the query time
This color was used to denote the TTL

3) Record Types

Record Type	Answer Output	What it is for
A	208.80.154.224	Maps the domain name of wikipedia.org to an IVP4 address
AAAA	2620:0:861:ed1a::1	Maps the domain name of wikipedia.org to an IVP6 address
MX	10 mx-in1001.wikimedia.org.	Shows the mail servers handling the domain
NS	ns0.wikimedia.org.	ID's the authoritative name servers that hold the DNS records for the domain
TXT	"v=spf1 include:_cidrs.wikimedia.org ~all"	Holds text data for domain ownership and verification records
SOA	ns0.wikimedia.org. hostmaster.wikimedia.org. 2026060420 43200 7200 1209600 3600	Stores core administrative details about the DNS zone, such as the primary name server and update timers.
CNAME	github.com.	An alias that redirects one domain to another
Reverse Lookup	dns.google.	Translates an IP address to it's hostname

4) +trace

```
; <<>> DiG 9.20.27 <<>> +trace wikipedia.org
;; global options: +cmd
.           419292  IN  NS  h.root-servers.net.
.           419292  IN  NS  b.root-servers.net.
.           419292  IN  NS  d.root-servers.net.
.           419292  IN  NS  c.root-servers.net.
.           419292  IN  NS  a.root-servers.net.
.           419292  IN  NS  f.root-servers.net.
.           419292  IN  NS  g.root-servers.net.
.           419292  IN  NS  e.root-servers.net.
.           419292  IN  NS  i.root-servers.net.
.           419292  IN  NS  j.root-servers.net.
.           419292  IN  NS  k.root-servers.net.
.           419292  IN  NS  l.root-servers.net.
.           419292  IN  NS  m.root-servers.net.
```

```
.           419292  IN  RRSIG   NS 8 0 518400 20260921170000
20260908160000 57780 . qgZvYF7MFKGa/
RbaWJEcVjbRccREcBVhelq5Fk+u+oJCakD4EPWcz38V
TWr+ufcrqWlzQsHhqB8XksN80F7ZUfR1NcRl0lP3ZamRpMpMWNGMzWFt
5pbTTTqHtxQDtoG3RVlXcWIYHrh2AoxIKbRHoVIo/XCui0evCWC637Kh
sax54IHbrNUY59IBC1wddIv/IqGTBm2wLDSBKe+o/M8qjyTo9WdsvSnS
jY3u42+1CRQTOjWv5ORO2HHOyC6mw0KfZW7lqBIEZN+V5LOdBhPiEsp+
odWCYlOF4ZpOmE7zt62S+y553s22fYuH8HWU184A2No9QY7QZirzmAg2
hfxz6w==
;; Received 1137 bytes from 128.153.5.248#53(128.153.5.248) in 8
ms

org.            172800  IN  NS  a0.org.afilias-nst.info.
org.            172800  IN  NS  a2.org.afilias-nst.info.
org.            172800  IN  NS  b0.org.afilias-nst.org.
org.            172800  IN  NS  b2.org.afilias-nst.org.
org.            172800  IN  NS  c0.org.afilias-nst.info.
org.            172800  IN  NS  d0.org.afilias-nst.org.
org.            86400   IN  DS  26974 8 2
4FEDE294C53F438A158C41D39489CD78A86BEB0D8A0AEAFF14745C0D
16E1DE32
org.            86400   IN  RRSIG   DS 8 1 86400 20260922170000
20260909160000 57780 .
imyoONliQu9zf7e35Fkdn0dvoxoW872A+ZMbxWKJBLfANFdUawZw3bI3
9dwIOlTiu490nFbr+LVqHVsY29syHs2FoKUVBjYgAgPFnmiFAsFpFEx5
Jr3R0qAFH1HOZ2D/ZJ04bCN3M+8W+BTsinpnpCrs55RDLO4YCfOR0Km5
Uu2491rRjGA/5RVwl7QUrlcTG/LSXZc6GaGaztrhLUCb8GARxRESSseG
eonpKC7e12xDvtM7fNRUFDlrRgXof4A6qp0h34sBqPMbusU88LO0sxlb
57ZOU0PuUVQWK9gUmBhgBKmsQeJXoi9RDIrhlAavXGihsR9mYVvIYV/l
o6gxVQ==
;; Received 821 bytes from 170.247.170.2#53(b.root-servers.net)
in 25 ms

wikipedia.org.      3600    IN  NS  ns1.wikimedia.org.
wikipedia.org.      3600    IN  NS  ns0.wikimedia.org.
wikipedia.org.      3600    IN  NS  ns2.wikimedia.org.
gdtpongmpok61u9lvnipqor8lra9l4t0.org. 3600 IN NSEC3 1 1 0
332539EE7F95C32A GDTREA8KMJ2RNEQEN4M2OGJ26KFSUKJ7 NS SOA RRSIG
DNSKEY NSEC3PARAM
gdtpongmpok61u9lvnipqor8lra9l4t0.org. 3600 IN RRSIG NSEC3 8 2
3600 20260930223145 20260909213145 25488 org.
oDxetwTo2O2xMhKRyVUBCkMBqI6ChqAaSh994DIO3DA5XDA+RLCiljhL
PvUo7d0FQplxvW8QZegF23+mbMl49u1xS/EZKh8j/sSdNuBK9K72IICr
pIZ48wyHfQJgKNkTrJ+MujWokmmJszY3CdSV3d2jEvUqUzirywuHOkV3 ud4=
tpebd49hd9o6as0iurn4p3ahh4smvg7m.org. 3600 IN NSEC3 1 1 0
332539EE7F95C32A TPEFKM9SVPQF3ET2HVHO7RSLFHNFTRRP NS DS RRSIG
```

```
tpebd49hd9o6as0iurn4p3ahh4smvg7m.org. 3600 IN RRSIG NSEC3 8 2
3600 20260930152312 20260909142312 25488 org.
DnfTNxKeE5y5FjuYLoNtMdht/B+ii/zpirK3r6rzgT8HfMpNa+yH5cQR
1XfaxLeXGknrkFpYo2qACjDkF6hZTH5vG2EEloqZ20ivSYvV/3/BXOh/
QEUbQY8bH4PX7caMi16EdokjTuZm2Do7pQacGyAG9X4pcuxOo18rKTs5 NtM=
;; Received 739 bytes from 199.19.56.1#53(a0.org.afilias-
nst.info) in 14 ms

wikipedia.org.      180 IN  A   208.80.154.224
;; Received 78 bytes from 208.80.154.238#53(ns0.wikimedia.org)
in 31 ms
```

(a) Root server that answered: b.root-servers.net (IP: 170.247.170.2)

(b) .org TLD servers referred to:

- a0.org.afilias-nst.info
- a2.org.afilias-nst.info
- b0.org.afilias-nst.org
- b2.org.afilias-nst.org
- c0.org.afilias-nst.info
- d0.org.afilias-nst.org

(c) Wikimedia's authoritative servers:

- ns0.wikimedia.org
- ns1.wikimedia.org
- ns2.wikimedia.org

Number of referrals: 3 referrals happened before receiving the final answer.

5) Caching

First Query Time: 39ms
Second Query Time: 30ms
Difference: 9ms
TTL: 1504s

If I had waited the full 1504 seconds for the TTL to expire then the next query would take about as long as the first one. This is because all of the cached information about the server would have been wiped.

6) Zone Transfer

I had 52 records return.

1) `contact.zonetransfer.me. 2592000 IN  TXT  "Remember to call or email Pippa on +44 123 4567890 or pippa@zonetransfer.me when making DNS changes”.`

This can be used to social engineer an attack to reconfigure the DNS in the attackers favor. It releases the PII of Pippa.

2) `cmdexec.zonetransfer.me. 300IN   TXT  "; ls”`

This is a Command injection

3) `vpn.zonetransfer.me. 4000 IN A 174.36.59.154`

VPN access gateway for remote entry into the internal network.

7) DNSSEC

```
flags: qr rd ra ad; QUERY: 1, ANSWER: 3, AUTHORITY: 0,
ADDITIONAL: 1

RRSIG record: cloudflare.com. 300 IN RRSIG A 13 2 300
20260912015839 20260909235839 34505 cloudflare.com.
15DyGpN+3+e5BYxFs4B0XXgQdlWOMSKIkju+pqwBXbQXMYJHZNnXZ6mp
yVfeelBtMo2HDPxncmljjxPjA5nbDQ==
```

The DS record in the com zone is holds a cryptographic hash of Cloudflare's Key Signing Key. This establishes a chain of trust from the parent TLD down to the cloud fare zone.

```
patrickfoulis@Patricks-MacBook-Air-4 ~ % dig @8.8.8.8 dnssec-
failed.org A

; <<>> DiG 9.20.27 <<>> @8.8.8.8 dnssec-failed.org A
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 32734
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0,
ADDITIONAL: 1
```

```
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
; EDE: 9 (DNSKEY Missing): (No DNSKEY matches DS RRs of dnssec-
failed.org)
;; QUESTION SECTION:
;dnssec-failed.org.    IN  A

;; Query time: 56 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sun Sep 13 14:53:37 EDT 2026
;; MSG SIZE  rcvd: 97

patrickfoulis@Patricks-MacBook-Air-4 ~ % dig @8.8.8.8 dnssec-
failed.org A +cd

; <<>> DiG 9.20.27 <<>> @8.8.8.8 dnssec-failed.org A +cd
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 20805
;; flags: qr rd ra cd; QUERY: 1, ANSWER: 1, AUTHORITY: 0,
ADDITIONAL: 1

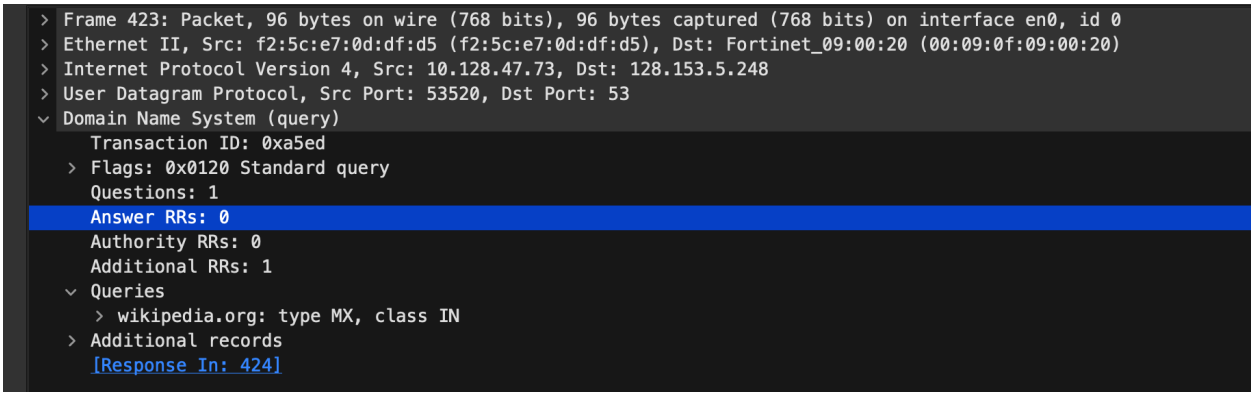
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;dnssec-failed.org.    IN  A

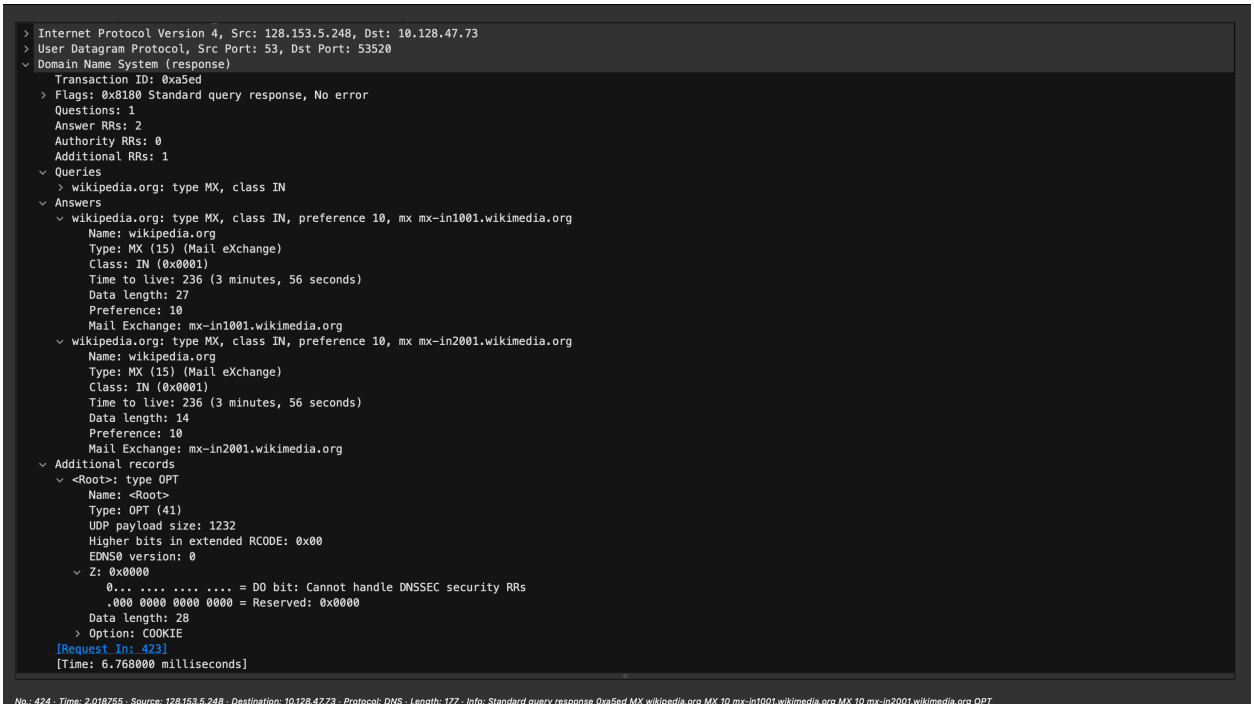
;; ANSWER SECTION:
dnssec-failed.org.  300 IN  A   96.99.227.255

;; Query time: 37 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sun Sep 13 14:53:53 EDT 2026
;; MSG SIZE  rcvd: 62
```

The +cd flag basically means turn of the DNSSEC validation. This is why the IP address was returned in the +cd flag instruction and it was not in the responded without an IP address. The default behavior of returning a SERVFAIL when validation fails is a security measure. This is because treating an unknown as a trusted response is dangerous.. So SERVFAIL is passed to tell the user that the response cannot be verified. This prevents against, man in the middle attacks, DNS spoofing, cache poisoning and many more attacks.

8) DNS On the Wire





The Transaction ID is a unique identifier used by the client to match an incoming DNS response to its corresponding original DNS query. The protocol to transport it that was used is UDP and sent to port 53 sent form 53520.

9) HTTP

The Request Line:

```
> GET / HTTP/1.1
```

The Three Request Headers:

```
 > Host: info.cern.ch
> User-Agent: curl/8.7.1
> Accept: */*
```

The Status Line:

```
< HTTP/1.1 200 OK
```

The Four Response Headers:

```
< Date: Sun, 13 Sep 2026 19:26:48 GMT
< Server: Apache
< Content-Type: text/html
< Content-Length: 646
```

The server header shows the exact software that the server is running. This was explored in Lab 2 as well, this is a weakness of information. If an attacker knows what the server is running, they can narrow down specific attack vectors to be used to exploit. If the CVE's are found for that specific software then all that is left to do is execute an attack.

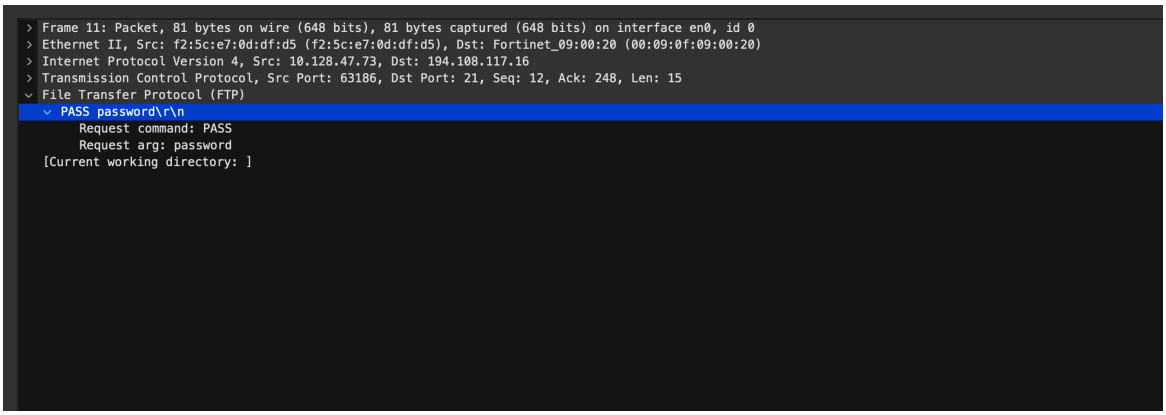
The Three Status Codes:
200
404
301
The Full Redirect Chain:
http://www.wikipedia.org/ -> https://www.wikipedia.org/
The difference in 301 and 302 are only in what happens after that specific request. 301 tells the browser to cache the redirect link for permanent use. Whereas 302 tells the browser to redirect the request to the original URL for future use.

10) HTTP Screenshot



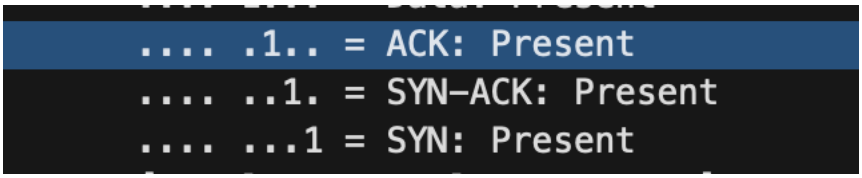
11) FTP Credentials

The server information given from the 220 greeting: test.rebex.net!\r\n



12) FTP Second Channel

The port number and the server the client told to use the data was port 1045 at 194.108.117.16. There was a visible TCP three way handshake on port 1045:



Thus proving that the second channel was opened. Passive mode was used here because the client requested it via `EPASV` allowing the client to initiate the data connection to the server's port 1045.

13) Encrypted Comparison

In the unencrypted FTP session, you could read application-layer plaintext data such as commands (`PASS`, `NLST`), credentials, directory contents, and welcome banners, which are completely encrypted as generic "Application Data" in the TLS session. However , from both sessions, you could read unencrypted network-level and transport-level metadata, including IP addresses, TCP/UDP port numbers, packet lengths, and handshake indicators.

14) RFC Questions

1)

- QDCOUNT
  - This specifies the number of entries in Question
- ANCOUNT
  - This specifies the number of reports in the answer section
- NSCOUNT
  - This specifies the number of reports in the authority section
- ARCOUNT
  - This specifies the number of records in the additional section

```
flags: qr rd ra cd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
```

2)

RD - Recursion Desired - “This bit may be set in a query and is copied into the response. If RD is set, it directs the name server to pursue the query recursively. Recursive query support is optional.”

RA -Recursion Available - “This be is set or cleared in a response, and denotes whether recursive query support is available in the name server.”

Both appeared because the client explicitly requested the server to resolve the full domain chain on its behalf (RD), and the local resolver accepted and supported that capability (RA).

3)

RFC 4035, Section 3.2.1
Reserving bits in early protocols allowed designers to introduce new security flags—like the AD bit for DNSSEC—without altering the fixed 12-byte header. Defining "Must be zero" ensured older legacy systems would safely ignore unused bits instead of misinterpreting them or crashing when new features are deployed later.

4)

Five status code classes (RFC 9110 Section 15.1):

1. 1xx: Informational
2. 2xx: Successful
3. 3xx: Redirection
4. 4xx: Client Error

5. 5xx: Server Error.

RFC 9110 Subsections for lab status codes:

1. 200 OK: Section 15.3.1
2. 301 Moved Permanently: Section 15.4.2
3. 404 Not Found: Section 15.5.5

5)

301 is a permanent redirect while 302 is a temporary redirect. This means that if you set a 301 accidentally you will permanently direct traffic away from your server. In this case you will have to wait for the clients to clear the cache before traffic will revert to your server. It is best to run a 302 until you are entirely sure a 301 is necessary.